

Multi-Prototype Compactness and Boundary-Aware Synthesis for Unsupervised Anomaly Detection

Kailun Liao Jianfeng Yang* Tao Tao Wenfei Wu Jiaming Jiang Jinsheng Xiao*

School of Electronic Information, Wuhan University

{liaokailun, yjf, xiaojs}@whu.edu.cn

Abstract

Unsupervised Anomaly Detection (UAD) is crucial for industrial quality control. Many existing embedding-based methods adopt a single-prototype assumption and learn, for example, a compact hypersphere to enclose all normal features. However, this strategy breaks down under intra-class variance caused by changes in illumination, pose, or texture. To cover all diverse normal samples, a single prototype may induce an overly loose decision boundary, making subtle anomalies hard to detect. To overcome this limitation, we propose PGBL (Prototype-Guided Boundary Learning), a framework that synergizes structured representation learning with targeted anomaly synthesis. First, the Multi-Prototype Compact Constraint (MPCC) module models the normal feature distribution as a mixture of multiple semantic prototypes, enabling tighter local representations for each normal sub-pattern. Second, instead of blind anomaly synthesis, the Boundary-Aware Anomaly Synthesis (BAAS) module generates pseudo-anomalies at the topological boundaries between MPCC clusters. Finally, a Discriminative Boundary Refiner (DBR) learns to shape the final decision surface by separating normal clusters from the synthesized anomalies. Extensive experiments on MVTec-AD, VisA and Real-IAD show that PGBL consistently outperforms prior methods in both anomaly detection and localization.

1. Introduction

Unsupervised Anomaly Detection (UAD) is crucial for various real-world vision applications, including defect detection [5], clinical image analysis [16], object detection [6], and video monitoring [36, 37]. The main challenge is training models solely on normal images to detect out-of-distribution patterns during inference.

Among various UAD methods, embedding-based methods [8, 18, 24] have achieved high detection performance. A

core idea of these methods is to map normal samples into a compact feature space, with the assumption that anomalies will fall outside this space. Some of these approaches rely on a single-prototype assumption [27, 29, 39], attempting to learn a compact unisphere, such as a minimum-volume hypersphere, to enclose all normal data points.

However, as shown in Fig. 1(a), this uni-modal assumption has limitations. Real-world normal data often exhibits high intra-class variance, forming a multi-modal distribution across diverse semantic states. First, modeling such diverse samples into a single global manifold forces the decision boundary to become overly general. This loose boundary encompasses inter-cluster gaps, leading to missed detections of weak defects. Second, optimizing solely under a single global constraint leaves the latent space internally unstructured, which risks feature confusion and representation collapse. While memory-bank methods [7, 26] preserve this variance by storing abundant features, they lack explicit geometric boundaries and incur high inference costs.

To address these issues, we propose the Multi-Prototype Compact Constraint (MPCC) to learn explicit boundaries while preserving local structures. Instead of forcing features into a single global manifold, MPCC captures intra-class variance by grouping patch features into distinct sub-clusters. As illustrated in Fig. 1(c), MPCC establishes a unified feature prototype space that couples global and local structures via one anchor prototype and multiple cluster prototypes. During training, each sample is mapped toward a shared anchor prototype and its nearest cluster prototype. This mixture prototype modeling drives the model to form multiple semantic and compact clusters that capture the intrinsic structure of normal samples.

While MPCC structures the normal feature space, the gaps between these prototype clusters remain undefined. To explicitly define these inter-cluster gaps as anomalous, we draw inspiration from feature-level anomaly synthesis methods. Existing strategies using Gaussian noise [20] generate anomalies with random directions and fixed scales, often producing invalid samples that overlap with normal clusters (Fig. 1(b)). While recent approaches [3, 4]

*Corresponding authors.

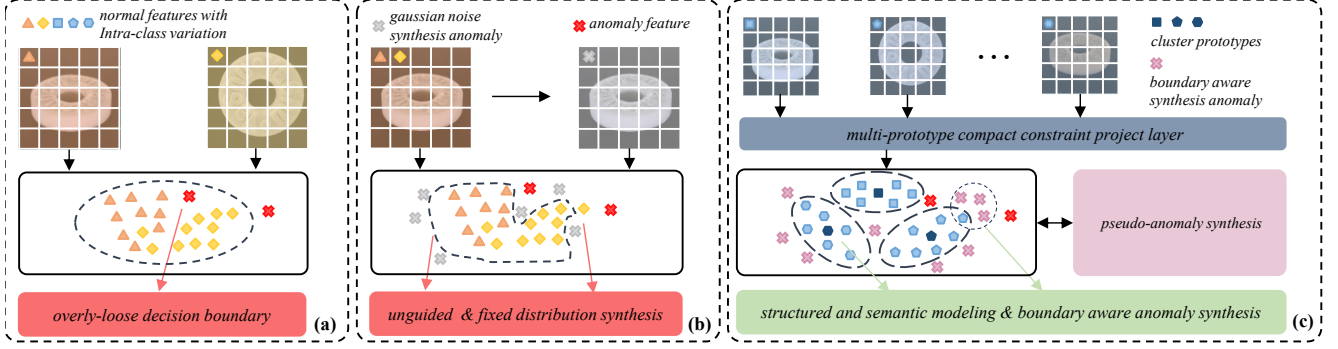


Figure 1. **A toy example to compare existing paradigms and our proposed framework** (a) Single prototype assumption: Overly-loose global boundaries due to multi-modal normal distributions, lacking local compactness. (b) Generating pseudo-anomalies blindly, or relying on parametric assumptions about normal distributions. (c) Our method utilizes Multi-Prototype Compact Constraint and Boundary-Aware Anomaly Synthesis to overcome these limitations of (a) and (b).

have achieved more targeted synthesis, they still rely on single-prototype hyperspheres or parametric assumptions about the normal distribution. Therefore, we propose the Boundary-Aware Anomaly Synthesis (BAAS) module. Unlike previous strategies, BAAS targets the latent boundary and works directly with MPCC. Using non-parametric density estimation, it identifies low-likelihood normal features to locate the topological boundaries between normal sub-clusters (Fig. 1(c)). By performing adaptive Gaussian sampling and rejecting high-likelihood candidates at these multi-prototype intersections, BAAS forces the network to learn a tight, non-linear decision boundary. Finally, the Discriminative Boundary Refiner (DBR) calibrates a precise margin by separating the synthesized anomalies from normal features. By pushing these hard negative samples away from the normal clusters, DBR tightens the decision boundary around the normal manifold without overfitting to specific anomaly types.

Together, these modules allow our PGBL framework to learn a feature space that is internally compact and externally well-separated. By combining structured representation learning with explicit boundary decisions, PGBL maintains high sensitivity to subtle anomalies while remaining robust to normal variations. Our main contributions are summarized as follows:

- We design the MPCC module to structure normal features into semantic sub-clusters, capturing intra-class variance without representation collapse.
- We propose the BAAS module, a novel distribution-free strategy that synergizes with MPCC and synthesizes targeted pseudo-anomalies by identifying boundary and inter-cluster gaps.
- Extensive experiments on MVTec-AD [2], VisA [45], and Real-IAD [34] demonstrate that PGBL achieves outstanding performance in both anomaly detection and localization tasks.

2. Related Work

Existing unsupervised anomaly detection methods can be generally divided into three paradigms: reconstruction-based, embedding-based, and data augmentation-based methods.

2.1. Reconstruction-based Methods

Reconstruction-based methods assume that models trained on normal data cannot accurately reconstruct abnormal patterns. Thus, anomalous regions can be highlighted by reconstruction errors. Recently, most of these methods are built upon AEs [15], GANs [10, 31], and diffusion models [42]. However, due to their powerful generalization capabilities, generative models sometimes reconstruct anomalous regions too well, rendering the residuals less obvious. To address this identity shortcut issue, some approaches force the model to recover normal patterns by applying inpainting or denoising mechanisms to masked or corrupted inputs [22, 40, 43, 44], while some methods strictly limit the model’s capacity to reconstruct unseen anomalies by utilizing discrete normal prototypes or minimalist architectures [11, 21, 23]. Despite these improvements, reconstruction-based methods still struggle with large anomalous regions and suffer from slow inference speeds.

2.2. Embedding-based Methods

Embedding-based methods extract features from pre-trained foundational models [14, 25] and detect anomalies by comparing the differences, such as distances, between the features of test and normal training samples. Depending on the stage of feature comparison, embedding-based methods can be roughly classified into several following categories. One-class Classification approaches [18, 29, 39] aim to find a hypersphere to separate abnormal features from normal ones. Knowledge Distillation approaches [8, 30, 38]

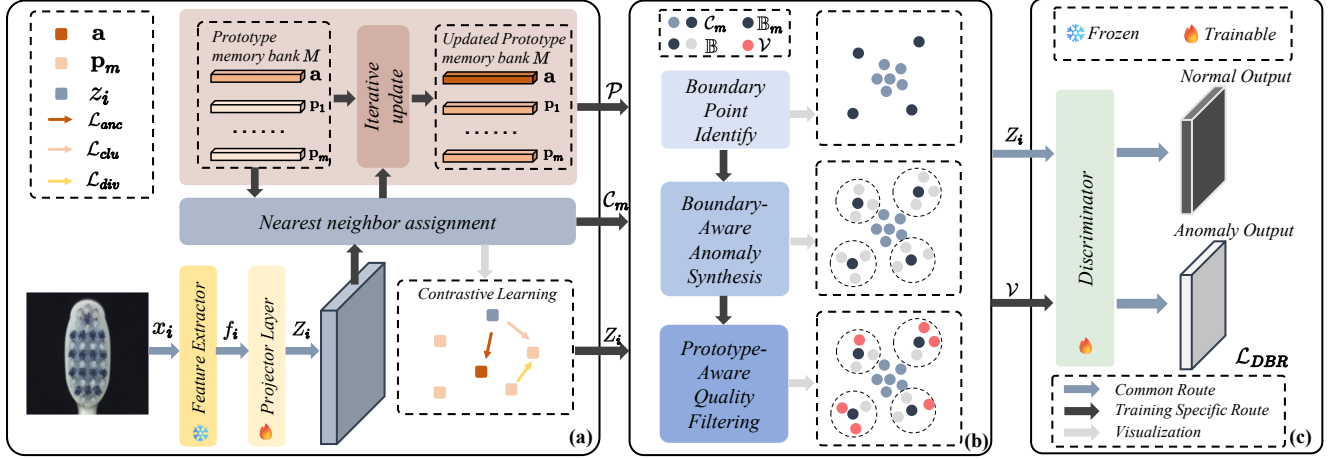


Figure 2. **Overview of the Proposed PGBL Framework.** (a) Multi-Prototype Compact Constraint (MPCC): employs a frozen feature extractor and a trainable projection head to map features into a structured embedding space, organizing them into compact clusters via a dual-level prototype system optimized by multi-level contrastive loss; (b) Boundary-Aware Anomaly Synthesis (BAAS): performs cluster-based and boundary-aware anomaly synthesis. It first executes boundary identification, followed by adaptive Gaussian sampling and prototype-aware quality filtering to generate pseudo-anomalies; (c) Discriminative Boundary Refiner (DBR): a discriminator trained to distinguish normal features from pseudo-anomalies.

identify anomalies by measuring the feature discrepancies between a pre-trained teacher and a student trained on normal data. Memory Bank approaches [7, 26, 35] store representative normal features to perform distance-based discrimination. Additionally, Normalizing Flow methods [28] estimate the probability density of normal features through bijective transformations. However, existing methods face a core trade-off: they either force diverse features into an overly generalized global manifold, failing to capture intra-class diversity, or overfit to normal data, yielding rigid decision boundaries. Consequently, they often trigger false positives on normal samples with rare variations or domain shifts.

2.3. Data Augmentation-based Methods

Data augmentation-based methods mitigate the lack of real anomalies by incorporating synthetic anomalies into the training frameworks. Image-level methods [19, 32, 41, 44] generate anomalies by pasting textures onto normal images or leveraging diffusion models. However, these approaches often lack realism, require auxiliary datasets, or incur heavy computational overhead.

Synthesizing anomalies directly in the feature space is more efficient, as it bypasses the need to repeatedly encode images. SimpleNet [20] pioneered this by injecting Gaussian noise into normal features. However, these synthetic features are mostly generated with fixed scales and random directions, which may fail to capture effective anomalous information. Recent advances have introduced directional constraints for anomaly synthesis [3], but are still limited by single-prototype hyperspheres. To overcome this, our

BAAS framework performs cluster-aware boundary identification, combining multi-cluster structures with adaptive directional guidance to synthesize effective feature-level anomalies.

3. Method

3.1. Overview

We present the pipeline of PGBL in Figure 2, featuring three main modules: MPCC, BAAS, and DBR. MPCC employs a frozen feature extractor $\mathcal{F}_\phi$ and a trainable projection head $\mathcal{G}$ to learn a compact multi-granularity feature space. BAAS identifies boundary points via k-NN distance and synthesizes pseudo-anomalies through Gaussian sampling. DBR is a classifier $\mathcal{D}$ trained to explicitly isolate synthetic anomalies, thereby calibrating a highly precise decision boundary. The framework is trained with $\mathcal{L}_{total} = \mathcal{L}_{MPCC} + \gamma \mathcal{L}_{DBR}$, where $\mathcal{L}_{MPCC}$ and $\mathcal{L}_{DBR}$ are losses from MPCC and DBR, respectively.

3.2. Pretrained Feature Extraction

Following prior work [26, 45], we employ a frozen ResNet-like backbone as our feature extractor. For any input image $x_i \in \mathbb{R}^{H_0 \times W_0 \times 3}$, we extract pretrained features $\phi_{i,j} \in \mathbb{R}^{H_j \times W_j \times C_j}$ from selected layers $j \in J$. To capture broader regional context and increase the receptive field, we aggregate neighborhood features using an adaptive average pooling layer, obtaining locally aware feature maps. Finally, we resize feature maps of all hierarchy levels to a common spatial resolution $H \times W$ via interpolation and concatenate them along the channel dimension.

The final feature map is denoted as $f_i \in \mathbb{R}^{H \times W \times C}$, where $C = \sum_{j \in J} C_j$ is the total channel capacity.

3.3. Multi-Prototype Compact Constraint

Some existing methods learn normal sample distributions by aligning to a single prototype [9, 18], but such approaches struggle when intra-class variance is present. To handle complex distributions, we propose Multi-Prototype Compact Constraint (MPCC), which models the normal feature distribution as a mixture of multiple compact sub-clusters. Optimized via a composite contrastive loss, MPCC employs a dual-level prototype system: an anchor prototype maintains class cohesion to prevent feature space fragmentation, while cluster prototypes learn discriminative representations that distinguish different sub-patterns.

MPCC maps the feature map $f_i \in \mathbb{R}^{H \times W \times C}$ to a new embedding space through a learnable projection head $\mathcal{G}$, which is implemented as a residual MLP block with equal input and output dimensions, resulting in $Z_i = \mathcal{G}(f_i) \in \mathbb{R}^{H \times W \times d}$, where $z_i^{h,w} \in \mathbb{R}^d$ denotes the projected feature vector at spatial location (h, w) .

Multi-level Prototype System. The prototype system consists of an anchor prototype $\mathbf{a} \in \mathbb{R}^d$ and a set of cluster prototypes $\mathcal{P} = \{\mathbf{p}_m \in \mathbb{R}^d\}_{m=1}^M$. We initialize $\mathbf{a}$ as the mean of projected features and $\mathcal{P}$ using K-Means clustering. We assign each feature vector $z_i^{h,w}$ to its nearest cluster prototype:

$$\hat{m}_i^{h,w} = \arg \min_m \|z_i^{h,w} - \mathbf{p}_m\|_2, \quad (1)$$

where $\hat{m}_i^{h,w}$ is the index of the closest prototype in $\mathcal{P}$. We denote $\mathcal{C}_m = \{z_i^{h,w} \mid \hat{m}_i^{h,w} = m\}$ as the cluster assigned to prototype $\mathbf{p}_m \in \mathcal{P}$. During training, we dynamically update the prototypes per batch by Exponential Moving Average [33]:

$$\mathbf{a} \leftarrow \alpha \mathbf{a} + (1 - \alpha) \bar{z}, \quad \mathbf{p}_m \leftarrow \beta \mathbf{p}_m + (1 - \beta) \bar{z}_m, \quad (2)$$

where $\bar{z}$ and $\bar{z}_m$ are the means of features over the current batch and features in $\mathcal{C}_m$, respectively, and α, β are momentum parameters.

Multi-level Contrastive Loss. To address the dynamic balance between cohesion and separation, we introduce a composite contrastive loss $\mathcal{L}_{MPCC}$ that combines three objectives. First, the Anchor Compact Loss provides global cohesion, ensuring the overall compactness and separability of the feature space. As defined in Eq. 3, it minimizes the distance between all projected features and the global anchor:

$$\mathcal{L}_{anc} = \frac{1}{N} \frac{1}{HW} \sum_{i=1}^N \sum_{h,w} \text{dist}(z_i^{h,w}, \mathbf{a}), \quad (3)$$

where N denotes the batch size, H, W represent the spatial dimensions of the feature map, and $\text{dist}(\cdot, \cdot)$ is a predefined distance metric, specifically the Euclidean distance.

Second, to handle high intra-class variance, the Cluster Compact Loss guides features toward their specific semantic cluster prototypes:

$$\mathcal{L}_{clu} = \frac{1}{N} \frac{1}{HW} \sum_{i=1}^N \sum_{h,w} \text{dist}(z_i^{h,w}, \mathbf{p}_{\hat{m}_i^{h,w}}). \quad (4)$$

Finally, to prevent prototype collapse, we apply a Prototype Diversity Loss that penalizes small distances between any pair of distinct prototypes:

$$\mathcal{L}_{div} = -\frac{1}{M} \sum_{m=1}^M \log \left(\min_{m' \neq m} \text{dist}(\mathbf{p}_m, \mathbf{p}_{m'}) \right). \quad (5)$$

To enable gradient backpropagation, we use the batch-wise cluster center $\bar{z}_m$ as a differentiable proxy for $\mathbf{p}_m$ in practice. The total MPCC loss is then formulated as the weighted sum of these three components:

$$\mathcal{L}_{MPCC} = \lambda_{anc} \mathcal{L}_{anc} + \lambda_{clu} \mathcal{L}_{clu} + \lambda_{div} \mathcal{L}_{div}, \quad (6)$$

where λ_{anc} , λ_{clu} , and λ_{div} are balancing hyperparameters.

3.4. Boundary-Aware Anomaly Synthesis

While MPCC enables the model to organize and represent the normal feature space through a multi-prototype system, MPCC itself only defines an implicit and ambiguous boundary. To address this, we draw inspiration from feature-level anomaly synthesis methods that explicitly calibrate the decision boundary by isolating near-distribution synthetic anomalies from normal representations. Existing feature-level synthesis methods rely on simple Gaussian noise with fixed variance [20], which is inefficient as many synthetic samples may fall far from the decision boundary or fall into normal sample clusters, or rely on distribution assumptions [4]. We propose Boundary-Aware Anomaly Synthesis (BAAS) to identify boundary regions using cluster assignments from MPCC and synthesize challenging anomalies specifically in these areas. BAAS consists of three steps: boundary point identification, anomaly synthesis, and quality filtering.

Boundary Point Identification. To identify the distribution boundaries, we employ k-NN distance as a non-parametric proxy for density. For each feature vector $z_i^{h,w}$ in cluster $\mathcal{C}_m$, we define its local sparsity score as the Euclidean distance to its k -th nearest neighbor:

$$\mathcal{S}_k(z_i^{h,w}) = \|z_i^{h,w} - \eta_k(z_i^{h,w}, \mathcal{C}_m)\|_2, \quad (7)$$

where $\eta_k(z_i^{h,w}, \mathcal{C}_m)$ denotes the k -th nearest feature vector to $z_i^{h,w}$ within $\mathcal{C}_m$. The boundary points $\mathbb{B}_m$ at the edge of the normal manifold are identified by selecting the top $P\%$ of features in $\mathcal{C}_m$ with the largest $\mathcal{S}_k$ scores. By aggregating

Algorithm 1 Boundary-Aware Anomaly Synthesis

Require: Projected feature maps $\{Z_i\}_{i=1}^N$, cluster assignments $\{\mathcal{C}_m\}_{m=1}^M$ from MPCC, cluster prototypes $\mathcal{P}$.

Ensure: Set of pseudo-anomalies $\mathcal{V}$.

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1:  $\mathbb{B} \leftarrow \emptyset, \mathcal{V} \leftarrow \emptyset$ 
2: for each cluster  $\mathcal{C}_m \in \{\mathcal{C}_m\}_{m=1}^M$  do
3:   Compute local sparsity score  $S_k(z)$  for all  $z \in \mathcal{C}_m$ 
   via Eq. 7.
4:    $\mathbb{B} \leftarrow \mathbb{B} \cup \mathbb{B}_m, \mathbb{B}_m = \{z_i^{h,w} \text{ in } \mathcal{C}_m \text{ with highest } S_k\}$ .
5: end for
6: for boundary point  $z_b \in \mathbb{B}$  do
7:   Sample candidate  $\tilde{v}$  via Eq. 8.
8:   if  $\tilde{v}$  satisfies the quality filtering condition in Eq. 9
   then
9:      $\mathcal{V} \leftarrow \mathcal{V} \cup \{\tilde{v}\}$ .
10:  end if
11: end for
12: return  $\mathcal{V}$ .
```

these local boundaries, the global boundary set is formed:
 $\mathbb{B} = \bigcup_{m=1}^M \mathbb{B}_m$.

Boundary-Aware Anomaly Synthesis. We synthesize candidate pseudo-anomalies by adding localized Gaussian noise to each boundary point $z_b \in \mathbb{B}$:

$$\tilde{v} = z_b + \epsilon, \epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (8)$$

where $\tilde{v}$ denotes a candidate pseudo-anomaly, and ϵ is the sampled Gaussian noise scaled by σ . This approach generates diverse anomalies at distribution boundaries without making assumptions about the overall normal sample distribution.

Prototype-Aware Quality Filtering. Unconstrained Gaussian sampling might generate pseudo-anomalies that fall too close to normal prototypes or cross into adjacent clusters. Let $d(x, \mathcal{P}) = \min_{\mathbf{p} \in \mathcal{P}} \|x - \mathbf{p}\|_2$ denote the distance from a feature x to its nearest prototype. For a boundary point $z_b \in \mathcal{C}_m$, we retain its synthetic candidate $\tilde{v}$ only if it lies further from any prototype than its originating boundary point z_b :

$$\mathcal{V} = \{\tilde{v} \mid d(\tilde{v}, \mathcal{P}) > \|z_b - \mathbf{p}_m\|_2\}. \quad (9)$$

where $\mathcal{P}$ represents the set of all cluster prototypes. This ensures synthetic anomalies $\mathcal{V}$ strictly reside in the low-density regions outside or between normal clusters. The overall process is summarized in Algorithm 1.

3.5. Discriminative Boundary Refiner

While MPCC structures a compact and semantic feature space, its decision boundary remains implicit. To explicitly calibrate this boundary, we propose the Discriminative Boundary Refiner (DBR). Inspired by [3], DBR employs

a lightweight MLP discriminator $\mathcal{D}$ trained to distinguish normal projected features z from synthetic anomalies $\tilde{v}$. The optimization objective is given by the standard Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{DBR} = \sum_z \text{BCE}(\mathcal{D}(z), 1) + \sum_{\tilde{v}} \text{BCE}(\mathcal{D}(\tilde{v}), 0), \quad (10)$$

where $z \in \{z_i^{h,w}\}$ denotes normal features and $\tilde{v} \in \mathcal{V}$ denotes synthetic anomalies. The loss gradients are backpropagated to both $\mathcal{D}$ and the projection head $\mathcal{G}$, creating a synergistic optimization that guides MPCC to produce a more separable feature space.

3.6. Training and Inference

Training. The entire framework is trained jointly by optimizing a composite loss function that combines the objectives of feature space structuring and boundary refinement:

$$\mathcal{L}_{total} = \mathcal{L}_{MPCC} + \gamma \mathcal{L}_{DBR}, \quad (11)$$

where γ is a hyperparameter that balances the two terms. During training, the projection head $\mathcal{G}$ and discriminator $\mathcal{D}$ are updated while the feature extractor $\mathcal{F}_\phi$ remains frozen.

Inference. During inference, only the trained feature extractor $\mathcal{F}_\phi$, projection head $\mathcal{G}$, and discriminator $\mathcal{D}$ are used. For a test image x , we extract features $f = \mathcal{F}_\phi(x)$ and project them to $z^{h,w} = \mathcal{G}(f^{h,w})$. Since the discriminator $\mathcal{D}$ is trained to output higher probabilities for normal features, we compute the pixel-level anomaly scores as $s^{h,w} = 1 - \mathcal{D}(z^{h,w})$, forming an anomaly map S . The map is upsampled for pixel-level localization via bilinear interpolation, and the image-level score $s_{img} = \max_{h,w} S_{h,w}$ is used for image-level detection.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate our method on three industrial anomaly detection datasets under single-class setting. **MVTec-AD** [2] contains 15 industrial product categories, divided into texture and object groups, with 3,629 normal images for training, and 1,982 anomalous images along with 498 normal images for testing. **Real-IAD** [34] is a large-scale real-world industrial anomaly detection dataset containing 30 different product categories with 151,230 images, featuring more realistic and diverse anomaly patterns. **VisA** [45] contains 12 categories of color industrial parts with a total of 10,821 images, covering multiple complex industrial scenarios including small-sized anomalies.

Implementation Details. Similar to [20], we use WideResNet50 pretrained on ImageNet as the backbone, and the feature extractor concatenates features from layers 2 and 3, resulting in a feature dimension of 1536. During training, input images are resized and center-cropped to 256×256 resolution. We use the StableAdamW optimizer to train the

Table 1. Comparison on MVTec-AD and VisA datasets. Best results in **bold**, second best underlined. Format: MVTec AD / VisA.

Metric	RD4AD [8]	SimpleNet [20]	DeSTSeg [43]	PatchCore [26]	NoCoAD [9]	UniAD [40]	MambaAD [12]	PGBL
I-AUROC	98.5 / 95.1	<u>99.6</u> / 95.4	98.6 / 90.4	99.2 / 92.8	97.7 / 94.4	95.8 / 90.6	98.7 / 95.3	99.8 / 98.0
I-AP	99.3 / 95.2	<u>99.8</u> / 94.8	99.5 / 91.1	99.8 / 91.4	99.7 / 94.9	98.9 / 92.3	99.6 / <u>95.3</u>	99.9 / 98.2
P-AUROC	97.5 / 98.6	98.1 / 98.1	98.1 / 98.5	98.1 / 98.5	<u>98.0</u> / 98.3	96.6 / 98.4	97.8 / 98.8	98.1 / <u>98.6</u>
P-AP	55.1 / 39.3	56.0 / 36.8	76.1 / <u>41.1</u>	57.2 / 40.8	60.5 / 36.8	45.8 / 31.6	57.2 / 40.4	<u>65.1</u> / 50.4
P-AUPRO	93.1 / 91.7	89.9 / 88.7	84.4 / 71.6	92.6 / 91.1	<u>94.1</u> / 94.9	88.8 / 85.3	94.0 / 91.9	95.0 / <u>92.6</u>

Table 2. Performance comparison on Real-IAD dataset. Best results in **bold**, second best underlined.

Metric	PaDiM [7]	PatchCore [26]	SoftPatch [17]	SimpleNet [20]	DeSTSeg [43]	UniAD [40]	RD [8]	MVAD [13]	PGBL
I-AUROC	80.3	89.4	89.4	88.5	86.9	82.9	87.1	<u>90.2</u>	91.5
P-AUPRO	81.8	91.5	90.8	84.6	81.5	86.1	93.8	<u>92.3</u>	<u>92.3</u>

projection head $\mathcal{G}$ and discriminator $\mathcal{D}$ with learning rates of 10^{-4} and 2×10^{-4} , respectively, and weight decay of 10^{-5} . Training consists of 200 epochs with a batch size of 8. The number of cluster prototypes M is set to 10. The anchor prototype momentum α and cluster prototype momentum β are set to 0.98 and 0.94, respectively. The loss weights are set to $\lambda_{anc} = 0.8$, $\lambda_{clu} = 1.0$, and $\lambda_{div} = 0.1$. For BAAS, boundary ratio P is set to 25, k-NN parameter k is set to 50 and variance σ^2 is set to 0.5. As a general guideline, these parameters are empirically scaled based on the target feature dimension. For DBR, the discriminator $\mathcal{D}$ is a 4-layer MLP and the weight γ is set to 1.0.

Evaluation Metrics. We employ the Area Under the Receiver Operating Characteristic Curve (AUROC) and Average Precision (AP) to evaluate the discriminative capability at both image and pixel levels, denoted as I-AUROC, P-AUROC, I-AP, and P-AP, respectively. Additionally, we calculate the Average Per-Region Overlap (AUPRO) at the pixel level, denoted as P-AUPRO, which is particularly sensitive to smaller-scale anomalies.

4.2. Comparative Experiments

To evaluate our proposed method, several typical and SOTA methods are employed in comparative experiments. NoCoAD [9], RD4AD [8], PatchCore [26] and PaDiM [7] employ an embedding-based framework. UniAD [40], RD [8], MVAD [13] and MambaAD [12] employ a reconstruction-based framework. SimpleNet [20] and DeSTSeg [43] employ an embedding-based framework integrated with anomaly synthesis.

Results on MVTec-AD and VisA. As shown in Table 1, PGBL achieves the best performance on both datasets. On MVTec-AD, PGBL achieves the best I-AUROC of 99.8%, I-AP of 99.9%, and P-AUPRO of 95.0%, with P-AUROC of 98.1% tying with SimpleNet, DeSTSeg, and PatchCore for the best result. On VisA, PGBL achieves the best I-AUROC of 98.0% and I-AP of 98.2%, while achieving P-

Table 3. Ablation study on MVTec-AD. Best results in **bold**.

MPCC	DBR	BAAS	I-AUROC	P-AUROC	P-AUPRO
			96.3	95.1	90.3
✓			97.0	97.4	93.1
✓	✓		99.2	98.0	94.3
✓	✓	✓	99.8	98.1	95.0

AUROC of 98.6% and P-AUPRO of 92.6%. The consistent superior performance across both datasets demonstrates the effectiveness and robustness of the proposed PGBL framework.

Results on Real-IAD. As shown in Table 2, PGBL achieves the best I-AUROC of 91.5% on Real-IAD, outperforming the second-best method MVAD (90.2%) by 1.3%. For P-AUPRO, PGBL achieves 92.3%, which is competitive with the top performers.

4.3. Ablation Studies

4.3.1. Component Effectiveness

As shown in Table 3, to evaluate the individual contributions of our proposed modules, we build our baseline similar to CFA [18], which represents the normal class with a single prototype derived from the average of all normal features, and defines the anomaly as the nearest-neighbor distance of a test feature to this prototype. The result in the second row demonstrates that the overall performance is improved by introducing the multi-prototype structure and the cluster prototype loss, as this improvement stems from MPCC’s ability to construct a more structured and semantically rich feature space. In the third row, we further introduce DBR as a discriminator and follow the strategy in SimpleNet [20], generating pseudo-anomalies by adding Gaussian noise to features. By deploying DBR as a binary classifier trained on Gaussian anomalies and normal features, the model is forced to delineate a tighter normality

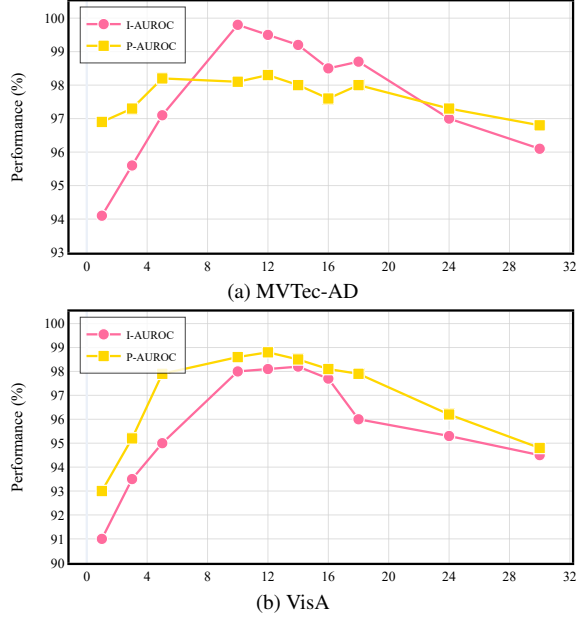


Figure 3. Sensitivity analysis of prototype number M on MVTec-AD and VisA datasets.

boundary. In the final row, we replace the anomaly synthesis method with BAAS, which proves that BAAS can synthesize higher-quality anomalies, thereby guiding DBR to further optimize the decision boundary and improve detection performance.

4.3.2. Parameter Analyses

Influence of the number of prototypes. We analyze the sensitivity of our approach to the hyperparameter M , which controls the number of cluster prototypes. Experiments were conducted on both MVTec-AD and VisA, sweeping M from 1 up to 30. According to the trends plotted in Fig. 3, an intermediate number of prototypes yields the highest detection accuracy. Extreme values degrade the results: too few clusters fail to capture data complexity, while too many introduce redundancy and overfit to the normal training distribution. The peak metrics are recorded at $M = 10$, where the model achieves 99.8% I-AUROC and 98.1% P-AUROC on MVTec-AD, as well as 98.0% I-AUROC and 98.6% P-AUROC on VisA. We therefore set $M = 10$ in all other experiments.

Influence of BAAS hyperparameters. As illustrated in Fig. 4, we conduct ablation studies on the MVTec-AD dataset to evaluate the impact of two parameters in BAAS: the neighborhood size k and the noise variance σ^2 . In Fig. 4a, we examine the effect of the neighborhood size used for local sparsity estimation by testing $k \in \{20, 40, 60, 80, 100, 120\}$. The results indicate that BAAS is robust and does not rely heavily on metic-

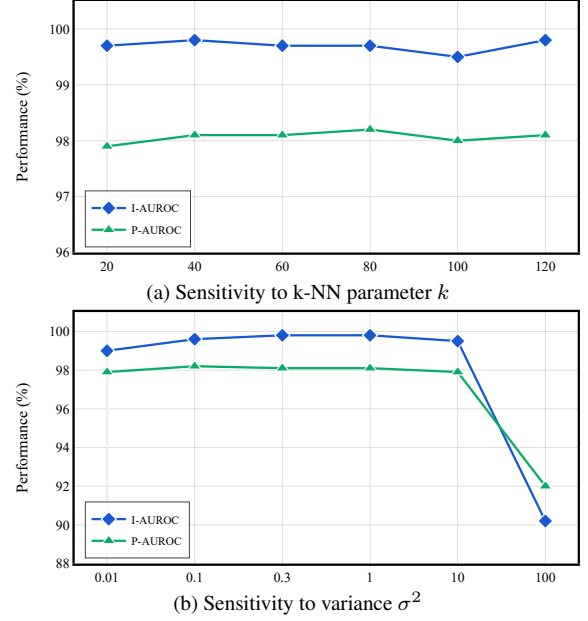


Figure 4. Sensitivity analysis of BAAS hyperparameters. (a) Impact of k for the k -NN distance. (b) Impact of variance σ^2 on pseudo-anomaly synthesis.

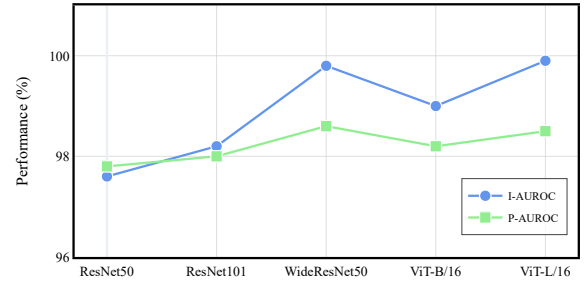


Figure 5. Effect of pretrained model.

ulously tuning the neighborhood size to identify boundary regions effectively. Furthermore, Fig. 4b presents the effect of the synthesis scale, where we evaluate $\sigma^2 \in \{0.01, 0.1, 0.3, 1, 10, 100\}$. Our method maintains consistent performance when a moderate variance is applied. However, excessively high σ^2 may cause injected noise to intrude into normal feature clusters, blurring the decision boundary and degrading performance.

Dependence on pretrained model. To explore the dependence on pretrained features, we evaluate our method with five backbones on MVTec-AD, including three ResNet-like and two ViT. As shown in Fig. 5, Performance remains high across backbones and WideResNet50 achieves the best P-AUROC of 98.6% and is only 0.1% behind ViT-L/16 on I-AUROC. We adopt WideResNet50 as the default backbone for its efficiency.

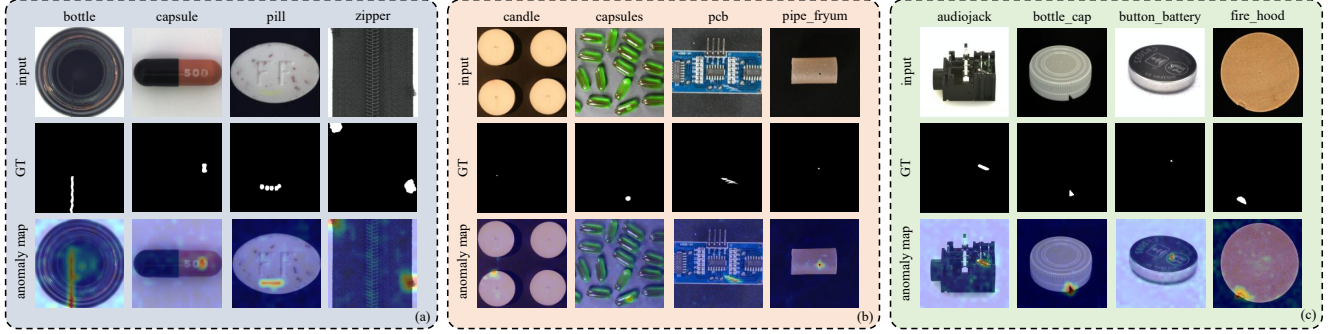


Figure 6. **Qualitative anomaly localization results** on MVTec-AD (a), VisA (b), and Real-IAD (c). The first row represents the input images, the second row represents the ground truth, and the third row displays the corresponding anomaly maps.

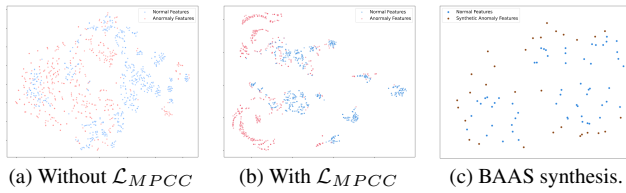


Figure 7. **Mechanistic visualizations** on the toothbrush category of MVTec AD via t-SNE. (a) and (b): normal and anomaly features. (c): synthesized anomaly features of BAAS within a batch.

4.4. Qualitative Results

Anomaly Localization. Fig. 6 visualizes the detection and localization results on MVTec-AD, VisA, and Real-IAD datasets. Qualitatively, multi-granularity compact prototypes help PGBL localize both fine-grained defects and large structural failures.

Mechanistic Visualization. To show the impact of components on feature space, we visualize normal and anomalous feature distributions using t-SNE. As shown in Fig. 7(a) and (b), under the constraint of $\mathcal{L}_{MPCC}$ and the projection layer, feature distributions become more compact, enabling better separation from anomaly samples. Fig. 7(c) shows that, within a batch, BAAS identifies boundary points of different clusters to generate pseudo-anomaly features.

Efficiency Comparison. We evaluate computational performance on an RTX 4090 GPU following the methodology of EfficientAD [1], with input size 256×256 . Latency is measured at batch size 1 and throughput at batch size 8. As shown in Table 4, PGBL achieves comparable inference efficiency to SimpleNet [20]. Although k-NN in BAAS adds additional computational overhead, this cost is mainly confined to the training phase. Consequently, inference remains low-latency and high-throughput, making PGBL suitable for real-world applications.

Impact of Random Initialization. Since prototypes are initialized via clustering and iteratively refined, we need to ensure they are not overly sensitive to the starting condi-

Table 4. **Efficiency comparison** on Real-IAD. Latency $\downarrow$ (ms) and Throughput $\uparrow$ (img/s) during training and inference.

Method	AUROC Image/Pixel	Latency Infer/Train	Throughput Infer/Train
SimpleNet [20]	88.5/84.6	6.77/7.83	215.08/185.7
PGBL	91.5/92.3	7.77/21.12	179.12/119.29

Table 5. **Sensitivity to random initialization** on MVTec-AD

Seed	I-AUROC	I-AP	P-AUROC	P-AP	P-AUPRO
0	99.8	99.9	98.1	65.1	95.1
1	99.8	99.7	98.1	65.0	95.0
2	99.7	99.8	98.0	64.9	95.0
3	99.8	99.8	98.0	64.8	94.9

tions. Table 5 reports the results over four random seeds. Small variance across all metrics shows that our training process is not sensitive to the initial seeds.

5. Conclusion

We address the problems of imprecise decision boundaries and insufficient modeling of intra-class differences in anomaly detection by proposing the PGBL framework, which consists of three core components: MPCC that structures the normal feature space into multiple semantic sub-clusters, BAAS that targets cluster margins to generate challenging synthetic anomalies, and DBR that explicitly calibrates a tight discriminative margin by isolating these synthesized hard negatives. Experiments on three datasets demonstrate that PGBL achieves state-of-the-art performance, validating the effectiveness of the prototype-guided boundary learning mechanism. However, our method is developed and evaluated solely under the single-class setting. Extending to multi-class or few-shot settings and introducing more flexible or online-updatable prototypes for evolving distributions are promising directions.

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